Differentiation: Inverse Hyperbolic Functions

Questions 1–27

Question 1. Show that

$$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}.$$

Question 2. Show that

$$\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}.$$

Question 3. Show that

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}.$$

Question 4. Differentiate each of the following:

a) $y = \operatorname{arsinh} 3x$
b) $y = \operatorname{arcosh} 2x$
c) $y = \operatorname{artanh} 5x$
d) $y = \operatorname{arsinh} \frac{1}{2}x$
e) $y = \operatorname{arcosh} \frac{x}{4}$
f) $y = \operatorname{artanh}\left(-\frac{x}{6}\right)$

Question 5. Differentiate each of the following:

a) $y = 2\,\text{arsinh}\,2x - 2\,\text{arcosh}\,4x$

b) $y = \frac{1}{3}\,\text{artanh}\,3x - \frac{2}{3}\,\text{artanh}\,6x$

Question 6. Find the equation of the tangent to the curve $y = 4\,\text{arsinh}\,x$ when $x = 2$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 7. Find the equation of the tangent to the curve $y = -6 \operatorname{arcosh} \frac{x}{2}$ when $x = 4$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 8. Find the equation of the tangent to the curve $y = 10 \operatorname{artanh} \frac{x}{5}$ when $x = 3$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 9. Determine whether the curve with equation

$$y = \operatorname{arsinh} 4x - \operatorname{arcosh} 2x$$

is convex (concave upwards) or concave (concave downwards) when $x = 2$.

Question 10. It is given that the curve with equation

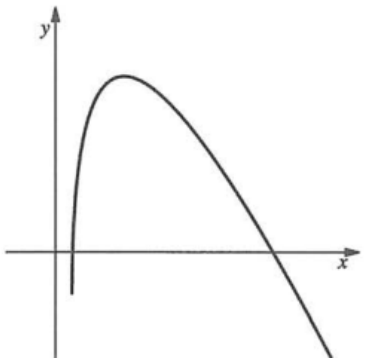
$$y = \sqrt{x^2 + 1} + \operatorname{arsinh} x$$

has a single stationary point. Find the exact coordinates of the stationary point.

Question 11. The graph of $y = f(x)$, where

$$f(x) = 12 \operatorname{arcosh} x - 3x, \qquad x \geq 1,$$

is shown below. Find the exact range of f, giving your answer in set notation.



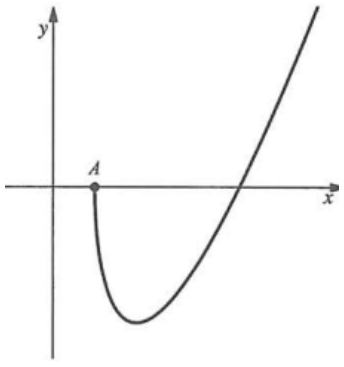
Question 12. The graph of $y = f(x)$ is shown below, where

$$f(x) = 2\sqrt{x^2 - 9} - 12 \operatorname{arcosh} \frac{x}{3}.$$

The point A lies on the curve and on the x-axis.

a) State the coordinates of the point A.

b) Find the range of f in set notation.



Question 13. It is given that the curve with equation

$$y = 4\,\text{artanh}\,x + \ln(x+1) - 6x$$

has two stationary points. It is also given that the curve has a single point of inflection.

a) Find the exact coordinates of both stationary points.

b) Find the exact x-coordinate of the point of inflection. Fully justify your answer.

Question 14. Differentiate each of the following:

a) $y = \text{arsinh}(\sqrt{x})$

b) $y = \text{arcosh}\left(\frac{1}{\sqrt{x}}\right)$

c) $y = \text{artanh}(\sqrt[3]{x})$

Question 15. It is given that $x > 0$.

a) Show that $\dfrac{d}{dx}\left(\operatorname{arsinh}\dfrac{1}{x}\right) = -\dfrac{1}{x\sqrt{x^2+1}}$.

b) A curve has equation $y = \sqrt{x^2+1} - \operatorname{arsinh}\frac{1}{x}$. Show that the curve has no stationary points.

Question 16.

a) Show that

$$\frac{d}{dx}\left(\operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}\right) = \frac{2}{\sqrt{4x^2+1}}.$$

b) A curve has equation

$$y = \sqrt{4x^2+1} - \operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}.$$

Show that the curve has a single stationary point and find its exact coordinates.

Question 17. It is given that

$$f(x) = \operatorname{arcosh} 2x + \operatorname{arcosh} \frac{2}{x}.$$

For $x > 0$, find the exact coordinates of the stationary point of the curve $y = f(x)$.

Question 18. Differentiate each of the following:

a) $y = \operatorname{arsinh}(\cosh x)$
b) $y = \operatorname{arcosh}(\tanh x)$
c) $y = \operatorname{artanh}(\sinh x)$
d) $y = \operatorname{arsinh}(\tanh x)$
e) $y = \operatorname{arcosh}(\sinh x)$
f) $y = \operatorname{artanh}(\cosh x)$

Question 19. Differentiate each of the following:

a) $y = x^2 \operatorname{arsinh} x$

b) $y = \sinh x \operatorname{arsinh} x$

c) $y = \cosh x \operatorname{arcosh} x$

d) $y = \operatorname{arsinh} x \operatorname{arcosh} x$

e) $y = \operatorname{artanh} 2x \operatorname{artanh} 4x$

Question 20. Differentiate each of the following:

a) $y = \dfrac{x}{\operatorname{arsinh} x}$

b) $y = \dfrac{\operatorname{arsinh} x}{\operatorname{arcosh} x}$

c) $y = \dfrac{\operatorname{arcosh} 3x}{\operatorname{arsinh} 6x}$

d) $y = \dfrac{\operatorname{arsinh} 4x}{\operatorname{artanh} 8x}$

Question 21. It is given that

$$y = \sinh(2\,\text{arcosh}\,x), \qquad x \geq 1.$$

Show that

$$(x^2 - 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 4y.$$

Question 22. It is given that

$$y = (\text{arsinh}\,4x)^2, \qquad x > 0.$$

Show that

$$(16x^2 + 1)\frac{d^2y}{dx^2} + 16x\frac{dy}{dx} = 32.$$

Question 23. Show that

$$\frac{d}{dx}(\operatorname{arcosech} x) = -\frac{1}{x\sqrt{x^2+1}}.$$

Question 24. Show that

$$\frac{d}{dx}(\operatorname{arcsech} x) = -\frac{1}{x\sqrt{1-x^2}}.$$

Question 25. Show that

$$\frac{d}{dx}(\operatorname{arcoth} x) = \frac{1}{1-x^2}.$$

Question 26. A curve has equation

$$y = x^2 \operatorname{arcosech} 2x, \qquad x > 0.$$

a) Differentiate y with respect to x.
b) Hence find the exact gradient of the curve when $x = \frac{1}{2}$.

Question 27. A curve has equation

$$y = \operatorname{arcsech} x^2 + \operatorname{arcoth} 3x,$$

where $-1 \leq x < -\frac{1}{3}$. Find the exact gradient of the curve when $x = -\frac{1}{2}$.